

Midterm 2 instructor guide

Slide route: Syllabus date: November 18, 2026. Proposed coverage: Lectures 5–7, with earlier prerequisites.

Session 10 | Midterm 2

Purpose: Second assessment session; no new content. The syllabus gives November 18. Calendar placement and exam duration require confirmation against the actual timetable.

Proposed blueprint

Area	Share	Evidence of learning
Multiple regression	35%	OVb sign, partial interpretation, reference categories.
Joint inference	30%	Joint null, restrictions, covariance-aware test.
Functional forms	35%	Log units, quadratic effect, interaction contrast.

The blueprint is a planning aid, not an approved exam. Announce final coverage in advance. If the course requires cumulative coverage for this midterm, adjust the blueprint explicitly.

100-minute slot assumption Ten minutes for instructions, 80 for the paper, ten for collection; replace with official timing if different. Confirm approved resources, room, and accessibility arrangements in advance.

What students should know A regression coefficient needs units and a conditioning statement. A joint test needs a joint null. An interaction effect needs a specified value of the interacting variable.

Instructor calibration examples

OVb An omitted positive determinant of Y negatively correlated with X creates negative bias in a simple-regression slope under the usual linear setup. Award credit for both links in the argument.

Restrictions Testing $\beta_1 = \beta_2$ is one restriction, not two. The variance of their estimated difference includes their covariance.

Interactions In $Y = 5 + 2X + 3D + 4XD + u$, the group contrast at $X = 2$ is $3 + 8 = 11$. The slope difference is 4. These answer different questions.

Logs A log-outcome coefficient .1 means an approximate 10% change, with exact exponentiated-index change $100(e^{.1} - 1) \simeq 10.52\%$.

After the paper

Record which errors concern algebra and which concern interpretation. Prepare a short return-to-teaching activity using a strong regression result with a doubtful causal story. This provides the transition to threats to identification.

Next session Lecture8v2.tex. Ask: even with correct standard errors and a flexible functional form, what can still go wrong?